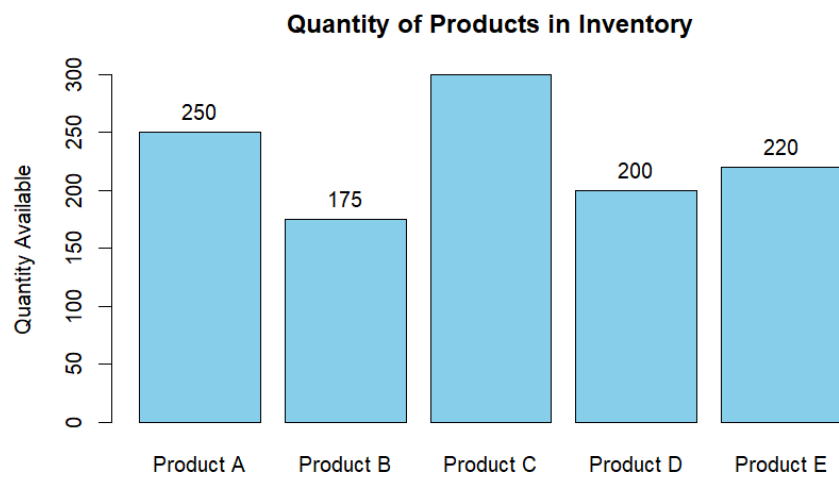


Set 4: Product Inventory Management

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

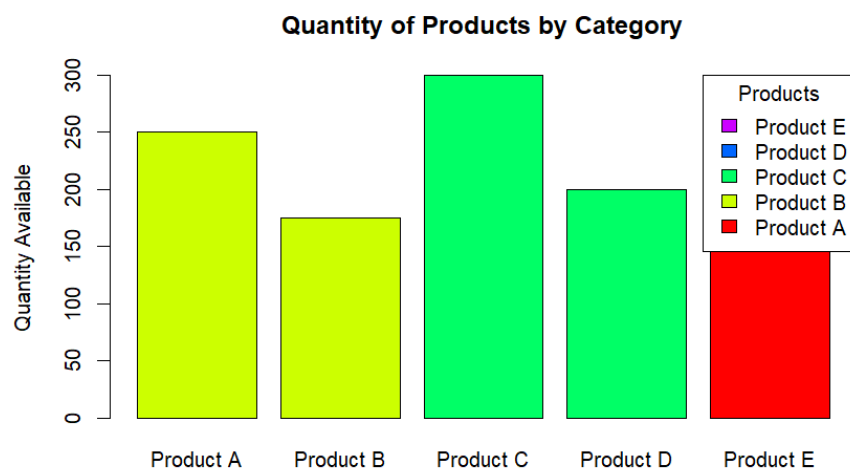
```
1 product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
2 quantity <- c(250, 175, 300, 200, 220)
3 barplot(quantity,
4         names.arg = product,
5         col = "skyblue",
6         border = "black",
7         main = "Quantity of Products in Inventory",
8         xlab = "Product Name",
9         ylab = "Quantity Available")
10 text(x = barplot(quantity, plot = FALSE),
11      y = quantity,
12      labels = quantity,
13      pos = 3)
```



```

1 product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
2 category <- c("Electronics", "Electronics", "Furniture", "Furniture", "Accessories")
3 quantity <- c(250, 175, 300, 200, 220)
4 data <- table(category, product)
5 data[,] <- 0
6 for(i in 1:length(product)) {
7   data[category[i], product[i]] <- quantity[i]
8 }
9 barplot(data,
10        col = rainbow(5),
11        main = "Quantity of Products by Category",
12        xlab = "Product Category",
13        ylab = "Quantity Available",
14        legend.text = colnames(data),
15        args.legend = list(title = "Products", x = "topright"))

```



```

1 product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
2 price <- c(50, 40, 70, 55, 60) # Assumed prices
3 quantity <- c(250, 175, 300, 200, 220)
4 plot(price,
5       quantity,
6       main = "Product Price vs Quantity Available",
7       xlab = "Product Price ($)",
8       ylab = "Quantity Available",
9       pch = 19,
10      col = "blue")
11 model <- lm(quantity ~ price)

```

